Ashwani Kumar Kamal

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Experience

Software Engineer Jul 2024 — Present
Schlumberger | Go, Java 22, Kubernetes, Postgres, Google Cloud Platform *Pune, India*

- Lead migration of core microservices from email to identity IDs, ensuring GDPR-compliance and reducing PII exposure.
- Migrated and tokenized databases containing **1M** records to adapt to new identity model by removing sensitivity from it.
- Improved identity lookups with SQL joins and implemented Redis cache layer reducing **P95** service latency by **20%**.
- Built a messaging library supporting both GCP Pub/Sub and Azure Service Bus, enabling multi-cloud deployments.
- Authored scripts used across **15+** services to backfill, transform, and tokenize records from Google Datastore to Postgres.
- Decommissioned unused task services, cutting infrastructure cost by **$630/month** and simplifying service topology.

Software Engineer Intern May 2023 — Jul 2023
Piramal Retail Finance | Java 17, Spring Boot, Temporal *Bengaluru, India*

- Designed proof of concept for in-house job scheduler for **15+** microservices to schedule functions in a durable workflow.
- Engineered microservice and client handling **10K+** scheduled tasks/day with configurable retry, payload, timeout options.

Open Source Contributions

Google Summer of Code, Lunatik - Lua in Kernel (github.com/luainkernel/lunatik) May 2026 — Sep 2026

- Designed a reusable eBPF layer for Lunatik, supporting Lua callbacks from eBPF across Traffic Control and `sched_ext`.
- Had **10+** GSoC patches merged upstream with close collaboration with the project maintainer and GSoC mentors.
- Implemented Lua APIs for eBPF maps, Linux Traffic Control and Extensible Scheduler including tests and demo scripts.
- Optimized object method dispatch with eager closure wrapping, reducing hot-path overhead by **53%** (**275ns** to **128ns**).
- Extended support for non-shared objects to bypass unnecessary locking and improve hot-path latency (**128ns** to **98ns**).

Contributor, etcd - Distributed key-value store used in k8s (github.com/etcd-io/etcd) Sep 2025 — Present

- Contributed to Auth API reliability by resolving retry issues and permission check when user sets the JSON web token.
- Eliminated duplicate delete/compact operations in robustness tests, reducing the cost of linearizability verification.

Projects

Maintainer, Goof (github.com/sneaky-potato/goof) Jan 2025 — Apr 2025

- Made a stack-based, statically typed programming language inspired by Forth targeting Linux **x86 64** architecture.
- Developed a Go compiler directly emitting assembly instructions, with lexical parsing, stack based IR and type inference.
- Generated statically linked ELF binaries and validated correctness with test programs (Game of Life, recursion, sorting).
- Support for **52** built-in words spanning control flow, procedures, stack manipulation, memory operations and system calls.

Competitions

Inter IIT Tech Meet 11.0 - Grow Simplee Challenge (Gold Medal) Jan 2023 — Feb 2023

- Deployed multiple HTTP and gRPC APIs to serve the frontend and optimization service with a **P95** latency of **120ms**.
- Implemented coordinate encoding using the Geohash feature of Redis reducing distance computation time by **30%**.

Education

Indian Institute of Technology Kharagpur Kharagpur, West Bengal
Bachelor of Technology in Computer Science and Engineering - CGPA 8.82 *2020 — 2024*

- Coursework: Operating Systems, Compilers, Computer Networks, Computer Organization and Architecture, Algorithms

Skills

- **Languages**: C, C++23, Go, Java (22/17), Lua, Python, TypeScript, Bash, SQL
- **Frameworks & Tools**: Spring Boot, Angular, Git/GitHub, Docker, Kubernetes, ArgoCD, GKE, PostgreSQL, Redis